A study on the adaptive relationship between green technology, green finance, and GDP

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Abstract

Advances in green technology and the allocation of green financial resources are regarded as two key forces driving the green transformation of the economy. However, most existing studies follow a single pathway centered on either technological innovation or financial support, and few compare their independent roles, joint information gains, and differences in model fit within a unified analytical framework. Based on 880 provincial-level observations, this study uses GDP as the dependent variable; constructs composite indices of green technology and green finance along with their 10 subvariables; trains and tests linear regression, ordinary least squares (OLS) regression, random forest, neural network, and support vector regression (SVR) models; and develops an adaptive multimodel matching rule. The results show that both green technology and green finance contain stable information for predicting GDP. Green finance primarily reflects capital allocation and project investment functions, whereas green technology more strongly reflects overall production efficiency and the level of technological development. Predictive performance improves substantially when both are included in the model, indicating complementarity between technological capability and the allocation of financial resources. The optimal model varies with the form of feature representation: SVR performs best when the two composite indices are used jointly, whereas random forest delivers the best overall performance when the 10 original subvariables are used.

Keywords: green technology; green finance; GDP; machine learning; adaptive model matching

1 Introduction

As the mode of economic growth shifts from factor-driven expansion toward quality improvement, maintaining stable growth under resource and environmental constraints has become a central issue in research on the green transition (Li and Cheng, 2022). Green technology can alter the technological foundation of economic growth by improving energy-use efficiency, optimizing production processes, and accumulating knowledge capital (Sun et al., 2026). Green finance, by contrast, channels financial resources toward low-carbon projects and green industries by adjusting the price of capital, access to financing, and the direction of

investment (Lin and Wang, 2025). The two act on the production possibility constraint and the resource allocation constraint, respectively, and may therefore jointly influence regional GDP (Wang and Wang, 2026). However, unified evidence on their relative importance, joint effects, and suitable identification models remains limited. It is therefore particularly important to determine how green technology and green finance are related to economic development separately, jointly, and under adaptive modeling conditions.

Existing research has developed along three relatively independent lines. First, studies of digital technology and green economic growth show that new technologies can promote green growth through industrial upgrading, pollution control, and firm innovation, although these effects may vary across regions. Methodologically, empirical research has expanded from single linear models to nonlinear threshold models, spatial econometric models, and efficiency measurement frameworks, thereby establishing a multidimensional testing system (Liu et al., 2022; Xu et al., 2023; Liang and Qiao, 2025; Yang and Kuang, 2025; Liu and Jiang, 2026). Baseline regressions generally confirm that green technology significantly promotes high-quality regional economic development. This effect is not constant, however, but is constrained by threshold conditions such as the stock of human capital and the degree of marketization; its driving force can be fully released only after these factors cross specific critical values. Spatial analyses further reveal that green technology not only increases local economic output but also improves growth performance in neighboring regions through technology diffusion and demonstration effects (Liu et al., 2022; Xu et al., 2023; Li et al., 2024; Alfalih and Hadj, 2024; Liang and Qiao, 2025; Qin et al., 2025; Alfalih, 2025; Yang and Kuang, 2025; Liu and Jiang, 2026). Heterogeneity tests indicate that the more developed eastern regions achieve growth mainly through industrial upgrading, whereas the less developed central and western regions rely more heavily on resource conservation and efficiency improvement. The magnitude and direction of the contribution of green technology therefore depend strongly on regional initial conditions and the institutional environment (Wu et al., 2022; Liu et al., 2022; Shang et al., 2024; Li et al., 2024; Liu and Jiang, 2026). Related studies generally use fixed-effects and mediation models to identify direct and indirect pathways. More specifically, most studies adopt panel fixed-effects models as their baseline specification and use mediation models to test transmission channels such as industrial upgrading, green total factor productivity, resource allocation efficiency, and energy intensity, thereby decomposing direct and indirect effects (Li et al., 2024; Qin et al., 2025; Liu and Jiang, 2026). Second, research on financial technology and corporate green technological innovation finds that financial service innovation can promote green technological innovation by easing financing constraints, attracting scientific and technological talent, and optimizing environmental regulation. With respect to financing constraints, green finance provides long-term funding for green research and development by reducing financing costs and broadening financing channels (Zhang, 2023; Zhang et al., 2026). With respect to talent agglomeration, green finance policies create incremental demand for green technology professionals, direct talent toward green industries, and generate knowledge spillovers (Li et al., 2026). With respect to environmental regulation, green finance provides regulated firms with funding for transformation and reduces their costs, thereby strengthening the innovation incentives created by environmental regulation

(Sun et al., 2026). Spatial Durbin models confirm positive spatial spillovers from these effects, difference-in-differences studies further verify the effectiveness of pilot-zone policies, and threshold regressions reveal that the effects are moderated by conditions such as the level of financial deepening and intellectual property protection (Zhang and He, 2026; Jia et al., 2025; Zhang, 2023; Li and Song, 2025; Qin et al., 2025). These effects are heterogeneous across regions, forms of ownership, and types of innovation. Regionally, talent agglomeration is more prominent in the east, whereas the central and western regions rely more on the easing of financing constraints. In terms of ownership, private firms are more sensitive to financing constraints. In terms of innovation type, incremental green patents have shorter commercialization cycles, whereas breakthrough green patents produce stronger long-term spillovers. Notably, green technology and green finance have complementary economic functions: green finance alleviates the financing constraints and risks faced by green projects, while the effectiveness of finance also depends on the reliability of the underlying technology projects (Wu et al., 2022; Zhang, 2023; Jia et al., 2025; Li and Song, 2025; Qin et al., 2025; Sun et al., 2025; Li et al., 2026; Zhang et al., 2026; Zhang and He, 2026). Third, the broader literature on innovation-led and finance-led growth generally holds that technological progress improves productivity. Super-efficiency slacks-based measure (SBM) models and the global Malmquist-Luenberger (GML) index are widely used to measure green technology efficiency and green total factor productivity. Researchers then use these measures as core explanatory variables and apply spatial Durbin models to examine spatial spillovers. The results show that technological progress not only raises local economic output but also improves growth performance in neighboring regions through technology diffusion and demonstration effects (Wu et al., 2022; Xu et al., 2023; Ye et al., 2023; Shang et al., 2024; Sun et al., 2025; Yang and Kuang, 2025; Liang and Qiao, 2025; Liu and Jiang, 2026; Madzianike et al., 2026). Tests of growth effects further confirm the robustness of the contribution of technological progress to productivity, with stronger gains in regions characterized by greater marketization and more abundant human capital. In terms of mechanisms, technological progress drives productivity growth primarily by improving factor-use efficiency, reducing energy consumption per unit of output, and shifting the production frontier outward, rather than simply expanding factor inputs (Wu et al., 2022; Liu and Jiang, 2026; Shang et al., 2024). Financial development also improves capital allocation. Instruments such as green credit and green securities improve the efficiency of capital formation, reduce information asymmetry, and redirect resources from inefficient sectors toward highly productive green industries. Improvements in capital allocation are reflected in the reorientation of credit toward firms with high research and development intensity and strong environmental performance, as well as in the greater ability of financial markets to price green assets. Spatially, the improvement in capital allocation brought about by financial development benefits not only the local region but also other regions through interregional capital flows and the cross-regional presence of financial institutions. Threshold analyses show that financial development improves capital allocation more strongly in regions with sound institutional environments and robust property-rights protection; otherwise, the positive effect may be weakened by the misallocation of financial resources (Zhang, 2023; Alfalih and Hadj, 2024; Jia et al., 2025; Li and Song, 2025; Qin et al., 2025;

Sun et al., 2025; Yen and Huyen, 2025; Zhang and He, 2026; Zhang et al., 2026). Nevertheless, three interrelated gaps remain. First, green technology and green finance are often examined separately, and joint comparisons within a unified framework are lacking. Second, existing studies rely predominantly on a single econometric model and seldom test whether the conclusions depend on the use of linear or nonlinear models. Third, studies based on composite indices may obscure differences among internal dimensions, whereas studies based on subvariables may overlook overall synergy. Accordingly, this study addresses three research questions: (1) What predictive relationships do green technology and green finance each have with GDP, and which internal dimensions contribute most? (2) Does their joint inclusion provide complementary information beyond that contained in single-factor models? (3) What adaptive model-matching results emerge for systems represented by composite indices and by original subvariables?

To address these questions, this study constructs indicator systems for green technology and green finance and compares the predictive performance and feature importance of multiple linear and nonlinear models. The study makes three contributions. First, it compares the independent roles of green technology and green finance and the information gain from their joint inclusion in predicting GDP. Second, it develops an adaptive model-matching mechanism based on a composite score from multiple evaluation metrics, thereby improving the ability of the analysis to accommodate nonlinearities and differences in feature structure. Third, it conducts a two-level analysis using both composite indices and their subvariables, establishing a correspondence between aggregate effects and internal structure.

2 Data and Methods

2.1 Data Sources and Preprocessing

This study examines 31 provincial-level administrative regions in China from 1992 to 2023, comprising 22 provinces, five autonomous regions, and four municipalities. The original consolidated dataset contains 992 province-year observations. After harmonizing and matching the green technology, green finance, and GDP indicators, 880 valid observations remained for model training and testing, including 704 observations in the training set and 176 in the test set, at a ratio of 8:2. To reduce the influence of cross-provincial differences in GDP scale and the right-skewed distribution on model estimation, $\log_2$GDP is used as the dependent variable. The original unit of GDP is CNY 100 billion.

The underlying data for green technology were obtained primarily from the China Statistical Yearbook, China Environmental Statistical Yearbook, China Agriculture Yearbook, China Agricultural Statistical Materials, China Rural Statistical Yearbook, China Energy Statistical Yearbook, China Water Resources Bulletin, satellite-derived data, and provincial and municipal statistical yearbooks. The underlying data for green finance were obtained primarily from the National Bureau of Statistics, Ministry of Science and Technology, People's Bank of China, Wind database, environmental exchanges, relevant statistical yearbooks, and environmental status bulletins, including the Almanac of China's Finance and Banking, China Insurance Yearbook, China Statistical Yearbook on Science and Technology, China Industry Statistical Yearbook, and

China Statistical Yearbook of the Tertiary Industry. Data from different sources were matched by province and year, and variable names, units, and statistical definitions were standardized.

Data processing was conducted at three levels. First, GDP was transformed using the base-2 logarithm. Second, green total factor productivity and its components were estimated within an input-desirable output-undesirable output framework, and the annually accumulated measure of green technological progress was retained. Third, the seven proportional indicators of green finance were positively normalized using min-max scaling, and objective weights were determined using the entropy method. Input features were standardized before model training. A random seed of 42 was used in models involving stochastic processes to ensure reproducibility.

Table 1. Variable Definitions and Measurement Methods

Category	Variable	Abbreviation	Measurement method	Economic interpretation
Dependent variable	Gross domestic product	GDP	Base-2 logarithm of original GDP	Level of economic development
Composite proxy for green technology	Cumulative value of the green technology progress index	CGTPI	Annual cumulative value of GTPI	Long-term stock of technological knowledge and cumulative effects
Composite green finance indicator	Green Finance Index	GFI	Entropy-weighted sum of seven green finance indicators	Overall level of green finance development
Green technology subvariables	Green total factor productivity	GTFP	GTFP / Global Malmquist-Luenberger (GML) index	Green productivity
	Green technology efficiency change index	GTEC	Efficiency-change component of the GML index	Improvement in technological efficiency (catch-up effect)
	Green technology progress change index	GTPCI	Technical-change component of the GML index	Current-period technological progress (innovation effect)
Green finance subvariables	Green credit	GCT	Total credit for environmental protection projects / total provincial credit	Allocation of green credit
	Green investment	GIT	Investment in environmental pollution control / GDP	Green capital input
	Green insurance	GIE	Environmental pollution liability insurance revenue / total premium revenue	Green risk protection
	Green bonds	GBD	Total green bond issuance / total bond issuance	Green direct financing
	Green support	GST	Government expenditure on environmental protection / general government budget expenditure	Government fiscal support for green development
	Green funds	GFD	Total market value of green funds / total market value of all funds	Allocation to green funds
	Green equity	GEY	Total trading volume of carbon emission rights, energy-use rights, and pollution discharge rights / total trading volume of the equity market	Depth of the green equity market

2.2 Index Construction

2.2.1 Measurement of GDP

This study uses regional GDP to measure provincial economic development. Given the substantial differences in GDP scale across provinces and the right-skewed distribution of the original values, $\log_2$GDP

is used as the dependent variable. The logarithmic transformation retains relative differences across regions while helping to reduce the influence of extreme scale values on parameter estimates and prediction errors.

2.2.2 Construction of the Green Technology Indicators

The level of green technology is measured using the Global Malmquist-Luenberger (GML) index, which incorporates undesirable outputs. The method uses the stock of fixed assets and average employment as inputs and regional GDP as the desirable output. Industrial solid waste, industrial wastewater, industrial waste gas, carbon dioxide emissions, and the annual mean concentration of fine particulate matter with an aerodynamic diameter of 2.5 micrometers or less (PM2.5) are included as undesirable outputs. The resulting unified framework captures intertemporal changes in green productivity while accounting for both economic output and resource and environmental constraints. The specific indicator system is presented in Table 2.

Table 2. Input-Output Indicator System for Measuring Green Technology

Category	Variable name	Unit	Data source
Input	Stock of fixed assets (estimated using the perpetual inventory method)	CNY 100 million	China Statistical Yearbook and provincial and municipal statistical yearbooks
	Average number of employed persons	10,000 persons	China Statistical Yearbook, China Agriculture Yearbook, China Agricultural Statistical Materials, and China Rural Statistical Yearbook
Desirable output	GDP	CNY 100 million	China Statistical Yearbook
Undesirable outputs	Industrial solid waste generated	10,000 metric tons	China Environmental Statistical Yearbook
	Industrial wastewater discharged	10,000 metric tons	China Environmental Statistical Yearbook
	Industrial waste gas emissions	100 million standard cubic meters	China Environmental Statistical Yearbook
	Carbon dioxide emissions	10,000 metric tons	China Energy Statistical Yearbook
	Annual mean PM2.5 concentration	micrograms per cubic meter	Satellite-derived data / China Environmental Statistical Yearbook

Let x_i, y_i, and b_i denote the input vector, desirable output, and undesirable output, respectively, for decision-making unit i, and let $D_0^t(\cdot)$ denote the directional distance function with period-t technology as the reference and (y, -b) as the direction vector. The GTEC, GTPCI, and GML are expressed as follows (Sutadian et al., 2016; Liu and Xin, 2019; Guo and Liu, 2020):

$$GTEC_i^{t,t+1} = \frac{1+D_0^t(x_i^t,y_i^t,b_i^t;y_i^t,-b_i^t)}{1+D_0^{t+1}(x_i^{t+1},y_i^{t+1},b_i^{t+1};y_i^{t+1},-b_i^{t+1})} \quad (1)$$

$$GTPI_i^{t,t+1} = \left[\frac{1+D_0^{t+1}(x_i^{t+1},y_i^{t+1},b_i^{t+1};y_i^{t+1},-b_i^{t+1})}{1+D_0^t(x_i^{t+1},y_i^{t+1},b_i^{t+1};y_i^{t+1},-b_i^{t+1})} \times \frac{1+D_0^{t+1}(x_i^t,y_i^t,b_i^t;y_i^t,-b_i^t)}{1+D_0^t(x_i^t,y_i^t,b_i^t;y_i^t,-b_i^t)}\right] \quad (2)$$

$$GML_i^{t,t+1} = GTEC_i^{t,t+1} \times GTPI_i^{t,t+1} \quad (3)$$

GTEC captures the catch-up effect of a decision-making unit relative to the production frontier, GTPCI captures the innovation effect represented by shifts in the production frontier, and GML is jointly determined by both components. To identify both current technological change and long-term technological accumulation, this study retains GTFP, GTEC, and GTPCI as three green technology subvariables. In the composite-indicator

setting, CGTPI is used as the composite proxy for green technology, termed the Green Innovation Index, to capture the long-term cumulative effect of technological progress.

2.2.3 Construction of the Green Finance Index

To comprehensively reflect the level of green finance development across provinces, this study constructs a composite evaluation system across seven dimensions: GCT, GIT, GIE, GBD, GST, GFD, and GEY. All seven indicators are positive proportional indicators; larger values indicate a higher level of development in the corresponding green finance channel. The definitions, data sources, and entropy-method weights of the indicators are reported in Table 3.

Table 3. Green Finance Indicator System, Measurement Methods, and Weights

Secondary indicator	Measurement method	Primary data source	Weight
GCT	Total credit for environmental protection projects / total provincial credit	Almanac of China's Finance and Banking; People's Bank of China	0.144883
GIT	Investment in environmental pollution control / GDP	China Environmental Statistical Yearbook	0.142964
GIE	Environmental pollution liability insurance revenue / total premium revenue	Almanac of China's Finance and Banking; China Insurance Yearbook	0.142503
GBD	Total green bond issuance / total bond issuance	People's Bank of China; Wind database	0.143462
GST	Government expenditure on environmental protection / general government budget expenditure	China Statistical Yearbook and provincial and municipal statistical yearbooks	0.140994
GFD	Total market value of green funds / total market value of all funds	People's Bank of China; Wind database	0.142211
GEY	(Carbon emission rights trading + energy-use rights trading + pollution discharge rights trading) / total equity-market trading volume	People's Bank of China; environmental exchanges	0.142982

To eliminate differences in dimensions and orders of magnitude, indicator X_{ij} for province-year observation i and indicator j is first positively normalized using min-max scaling (Huang et al., 2023; Liu et al., 2025; Ma et al., 2025):

$$X'_{ij} = [X_{ij} - min(X_j)]/[max(X_j) - min(X_j)] \quad (4)$$

Next, the normalized values are used to calculate the indicator share p_{ij}, entropy value e_j, coefficient of variation d_j, and objective weight ω_j:

$$p_{ij} = X'_{ij}/\sum_{i=1}^{n} X'_{ij}; \quad \mathrm{e_j} = -(1/\ln n)\sum_{i=1}^{n} p_{ij}\ln(p_{ij}) \quad (5)$$

$$d_j = 1 - e_j; \ \ \omega_j = d_j/\sum_{j=1}^{m} d_j, \text{and} \sum_{j=1}^{m} \omega_j = 1 \quad (6)$$

When $p_{ij} = 0 = 0$, define $p_{ij}\ln(p_{ij}) = 0$. The greater the dispersion of an indicator, the lower its entropy value and the higher its information utility, and hence the greater its corresponding weight. Finally, the Green Finance Index is expressed as follows:

$$GFI_i = \sum_{j=1}^{m} \omega_j X'_{ij}, \ m = 7 \quad (7)$$

The theoretical value of the Green Finance Index (GFI) lies between 0 and 1. A larger value indicates a higher level of green financial resource allocation and market development in the province.

2.3 Experimental Design

2.3.1 Feature Configurations and Experimental Groups

To evaluate the independent predictive power of green technology and green finance, the information gain from their joint inclusion, and differences in model fit across indicator levels, six feature configurations are specified. The design first compares the separate roles of the two categories of green factors, then tests the incremental explanatory power obtained when both are included in the model, and finally compares the predictive performance of composite proxy indicators with that of the original subvariables.

Table 4. Experimental Groups and Feature Configurations

Experimental group	Input features	Research purpose
Experiment 1	Seven green finance subvariables	Identify the predictive contribution of internal green finance channels
Experiment 2	Three green technology subvariables	Identify current efficiency, technological progress, and cumulative effects
Experiment 3	CGTPI	Test the predictive ability of the composite proxy for green technology
Experiment 4	Green Finance Index (GFI)	Test the predictive ability of the composite green finance index
Experiment 5	Green Innovation Index + GFI	Test the joint information gain from technological accumulation and financial allocation
Experiment 6	All 10 green-factor subvariables	Compare nonlinearities and interaction structures among disaggregated indicators

2.3.2 Relationship Models and Parameter Settings

This study compares the fit and generalization error of five models on the same training and test sets: linear regression, ordinary least squares (OLS) regression, random forest regression, support vector regression (SVR), and neural network regression (Hastie et al., 2009; Terkes et al., 2026). The linear models provide benchmark directions and interpretations of marginal associations, whereas the nonlinear models identify thresholds, interactions, and complex functional relationships.

(1) Linear regression. Let y_i denote $\log_2$GDP for observation i, x_{ij} denote green technology or green finance feature j, and p denote the number of features. The baseline linear relationship is:

$$y_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} + \varepsilon_i \quad (8)$$

where β_0 is the intercept, β_j is the feature coefficient, and ε_i is the random error. In the linear model, the standardized absolute coefficient, $|\beta_j|$, is used as an approximate measure of relative feature importance.

(2) Ordinary least squares regression. OLS estimates the parameters by minimizing the sum of squared residuals:

$$\hat{\beta} = arg\ min\beta\ \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})^2 \quad (9)$$

OLS provides clear coefficient directions and marginal associations but is sensitive to nonlinearities, outliers, multicollinearity, and interactions among variables.

(3) Random forest regression. Random forest reduces the variance of an individual tree by averaging the predictions of B regression trees:

$$\hat{f}RF(x) = (1/B)\sum_{\beta=1}^{B} T_{\beta}(x) \quad (10)$$

The model uses B = 500 trees, a maximum tree depth of 10, a minimum of five samples to split an internal node, and a minimum of three samples per leaf. At each split, the number of randomly selected features equals the square root of the total number of features. The random seed is fixed at 42. Normalized feature importance is used to measure the relative predictive contribution of each variable.

(4) Support vector regression. SVR finds the flattest possible regression function in a high-dimensional feature space while permitting errors within an epsilon-insensitive band:

$$min\ 1/2\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \quad (11)$$

$$s.t.\ y_i - w^T\varphi(x_i) - b \le \varepsilon + \xi_i;\ w^T\varphi(x_i) + b - y_i \le \varepsilon + \xi_i^*;\ \xi_i, \xi_i^* \ge 0 \quad (12)$$

A Gaussian radial basis function (RBF) kernel is used, with penalty parameter C = 10, insensitivity coefficient ε = 0.1, and the kernel coefficient γ determined automatically using the scale strategy.

(5) Neural network regression. The neural network approximates complex functions through multilayer nonlinear mappings. The hidden representation at layer l and the output layer can be written as:

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)}),\ \hat{y} = W^{(L)}h^{(L-1)} + b^{(L)} \quad (13)$$

The input-layer dimension adapts to the six feature configurations. The two hidden layers contain 100 and 50 neurons, respectively, and use the rectified linear unit (ReLU) activation function; the output layer uses linear activation. The model uses the Adam optimizer, a batch size of 16, a maximum of 1,000 iterations, and an L2 regularization coefficient of α = 0.01. A random 10% of the training set is used as the validation set, early stopping is enabled, and the parameters associated with the minimum validation loss are restored after training. The random seed is fixed at 42.

2.3.3 Evaluation Metrics and Adaptive Model Matching

To avoid bias from selecting models on the basis of a single metric, this study evaluates test-set performance using the coefficient of determination (R^2), mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE):

$$R^2 = 1 - [\sum_i(y_i - \hat{y}_i)^2]/[\sum_i(y_i - \bar{y})^2]。 \quad (14)$$

$$MSE = (1/n)\sum_i(y_i - \hat{y}_i)^2;\ RMSE = \sqrt{MSE};\ MAE = (1/n)\sum_i|y_i - \hat{y}_i|。 \quad (15)$$

A larger R^2 indicates greater explanatory power, whereas smaller values of RMSE, MSE, and MAE indicate lower prediction error. For the five models under each feature configuration, R^2 is first positively normalized using min-max scaling, while the three error metrics are inversely normalized. An equally weighted composite score is then calculated:

$$Score_m = (\tilde{R}_m^2 + \widetilde{RMSE}_m + \widetilde{MSE}_m + \widetilde{MAE}_m)/4, m^* = arg\ max_m Score_m。 \quad (16)$$

A larger $Score_m$ indicates better overall model performance, and m* denotes the adaptively selected optimal model for the current feature configuration. This rule allows model selection to account

simultaneously for explanatory power and multiple error scales and to adjust as the feature structure changes between composite indicators and original subvariables.

3 Results and Analysis

3.1 Predictive Relationships of Green Technology and Green Finance with GDP

Using the green technology subvariables, green finance subvariables, and their respective composite indices as independent variables and GDP as the dependent variable, the predictive relationships are analyzed with linear regression, random forest, neural network, SVR, and OLS models. The results are shown in Figure 1.

The single-factor models show that both green technology and green finance contain information for predicting GDP, but the predictive ability of the indicator systems differs markedly. In the model with seven green finance subvariables, random forest achieves a test-set R^2 of 0.5765, with RMSE, MSE, and MAE values of 1.2369, 1.5300, and 0.9672, respectively, outperforming the other models overall. GIT ranks first or within the most important group in all five models, with relative importance values of 0.3733, 0.4516, and 0.3726 in linear regression, neural network, and OLS, respectively; both random forest and SVR also rank it first. GFD and GEY make relatively large contributions in the linear models, whereas the importance of variables such as GIE and GBD increases in random forest and SVR. This suggests that different green finance instruments may jointly provide information related to economic scale through nonlinear combinations.

When only the three green technology subvariables, namely GTFP, GTEC, and GTPCI, are included, the overall predictive ability of all models is substantially weaker. Random forest produces a test-set R^2 of 0.0601 and RMSE and MSE values of 1.8427 and 3.3955, respectively, making it the best-performing model overall in this setting. SVR yields the lowest test-set MAE, at 1.4952, but its R^2 is only 0.0372. These findings show that the three current-period green technology indicators explain only a small proportion of the variation in GDP. Current changes in technological efficiency and technological progress alone cannot adequately capture the relationship between green technology and economic scale.

The models also differ considerably in how they identify the green technology variables. Linear regression places greater emphasis on GTEC and GTPCI, with importance values of 0.4326 and 0.3359, respectively. Random forest, neural network, SVR, and OLS all identify GTFP as the most important variable, assigning importance values of 0.4117, 0.8604, 0.6465, and 0.5187, respectively. Feature importance is particularly concentrated on GTFP in the neural network and SVR models, while the importance of GTPCI in SVR is only 0.0006. This indicates that the nonlinear models extract predictive information primarily from the level of green productivity, whereas current-period changes in technological progress make a relatively limited independent contribution to GDP prediction.

When composite indices are used, the best test-set result for the Green Innovation Index is obtained with the neural network, with an R^2 of 0.5761; the best result for the GFI is obtained with SVR, with an R^2 of 0.5611. The Green Innovation Index predicts substantially better than a model that directly includes the three

green technology subvariables. This suggests that the composite index may reduce fluctuations and measurement noise in individual indicators through information aggregation, thereby capturing the overall characteristics of green technological development more effectively. When modeled separately, the Green Innovation Index and GFI have similar explanatory power, indicating that both are important for predicting GDP. Their complementarity and joint predictive performance nevertheless require further examination in a joint model.

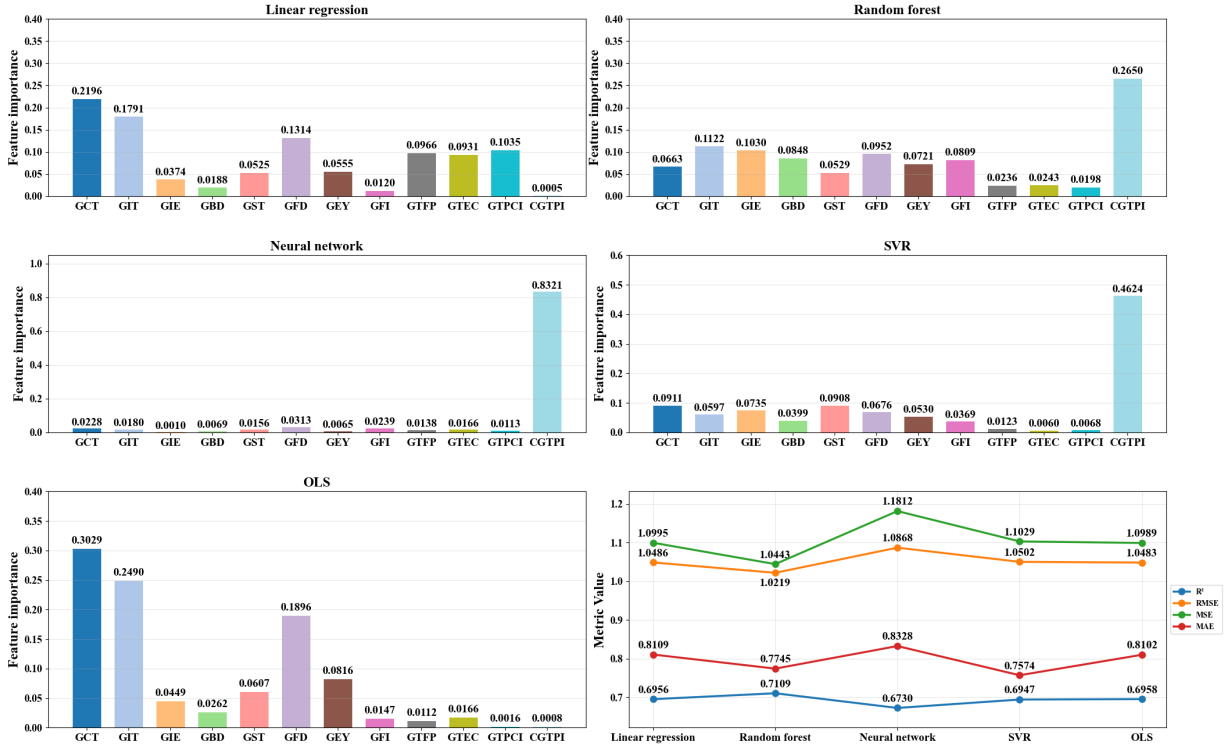


Figure 1. Prediction of Log_2GDP by green technology/finance subvariables and composite indices across five models.

3.2 Joint Information Gain from Green Technology and Green Finance

The predictive relationship obtained from the five models when green technology and green finance are jointly used as independent variables and GDP is used as the dependent variable is shown in Figure 2.

When the Green Innovation Index and GFI are included simultaneously, test-set performance improves markedly across all five models. SVR achieves the highest R^2 (0.7331) and the lowest RMSE (0.9820), MSE (0.9643), and MAE (0.7147). Relative to the best green technology single-factor model, the R^2 of the joint model increases by 0.1570; relative to the best green finance single-factor model, it increases by 0.1720. These gains indicate that technological accumulation and financial allocation do not merely provide duplicate measures of the same development condition; rather, they provide complementary information about the productivity foundation and capital support, respectively.

Feature importance in the joint-index model also reveals substantial differences in model interpretation. Linear regression and OLS estimate the importance of GFI at 0.9751 and 0.9770, respectively, indicating that the linear models are almost entirely dominated by the financial variable. Random forest, neural network, and SVR, by contrast, allocate approximately 0.48-0.52 of feature importance to both GFI and CGTPI. Thus, relying solely on linear coefficients may understate the nonlinear role of cumulative green technological progress. The nonlinear models provide stronger support for the complementary interpretation that green finance supplies resources while green technology creates productive capacity.

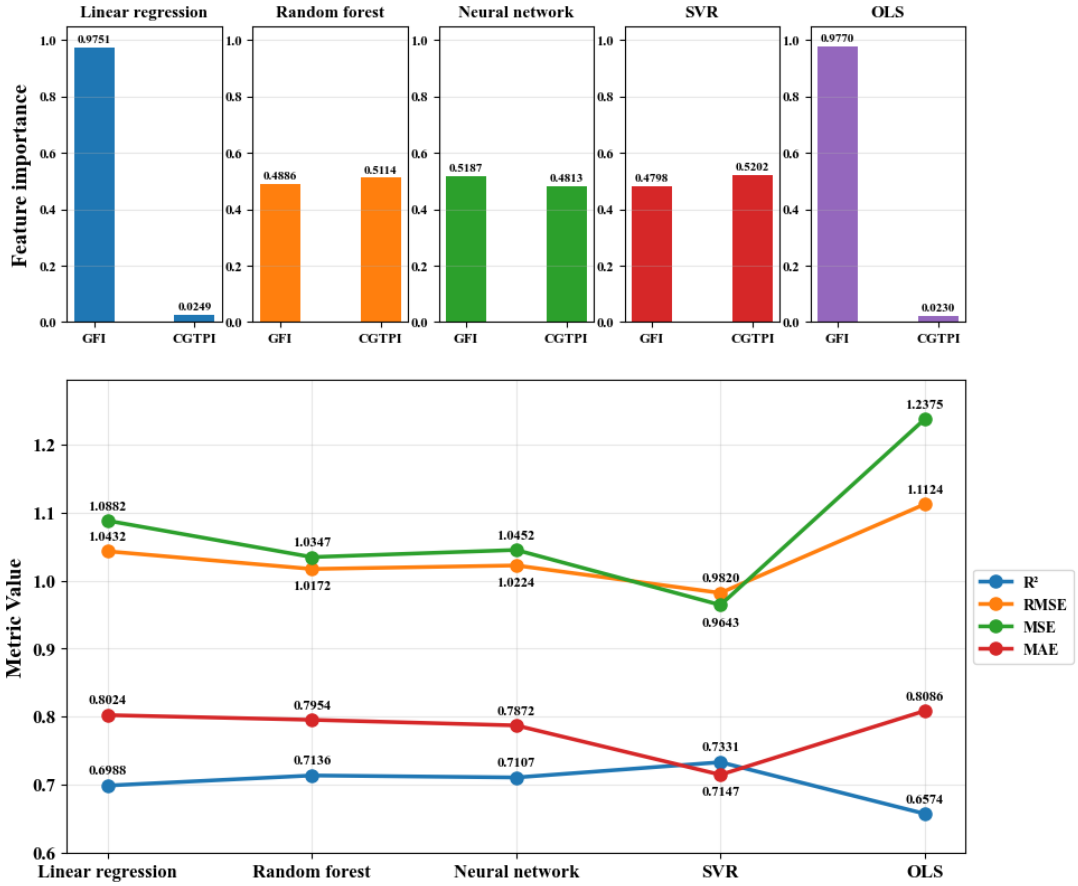


Figure 2. Prediction of Log_2GDP by CGTPI and GFI using five models.

3.3 Nonlinear Relationships Between Green Factors and GDP

The joint predictive relationship between green factors and GDP is analyzed using five models, with GTFP, GTEC, GTPCI, GCT, GIT, GIE, GBD, GST, GFD, and GEY as the 10 independent subvariables and GDP (log_2GDP) as the dependent variable. The results are shown in Figure 3.

The optimal model changes with the indicator level, feature dimension, and method of combining variables. Based on the joint assessment of test-set R^2 and the error metrics, random forest performs best in both the green finance subvariable setting and the green technology subvariable setting; the neural network

performs best for the Green Innovation Index; SVR performs best for both the GFI and the joint-index setting; and random forest again achieves the best overall result when all 10 subvariables are included. These findings show that indicator structures differ in their linear relationships, nonlinear relationships, and degree of interaction. Model selection therefore cannot rely on a single uniform functional form but should instead reflect feature dimensionality and data structure.

When all 10 subvariables are included simultaneously, random forest achieves a test-set R^2 of 0.5967, with RMSE, MSE, and MAE values of 1.2071, 1.4570, and 0.9233, respectively. It outperforms the other models on all four test metrics and is therefore the best model overall in this setting. SVR achieves a test-set R^2 of 0.5226, with RMSE and MAE values of 1.3133 and 1.0017, respectively; the test-set R^2 of the neural network is 0.5166. The training-set R^2 of random forest is 0.8032, indicating a gap between its training- and test-set performance. Although the model captures complex relationships in the training sample relatively well, its out-of-sample generalization stability requires attention.

Random forest feature importance ranks GIE, GIT, GFD, GBD, and GEY as the five most important variables, with values of 0.1589, 0.1580, 0.1485, 0.1329, and 0.1224, respectively. The importance values of all three green technology variables are below 0.05: GTEC, GTFP, and GTPCI have importance values of 0.0462, 0.0458, and 0.0298, respectively. These results indicate that, within the nonlinear joint relationships identified by random forest, most of the information used to predict GDP comes from combinations of green finance instruments, while the three current-period green technology variables provide relatively little marginal information.

The models differ to some extent in their identification of the core variables. Linear regression and OLS mainly emphasize GIT, GCT, and GFD. Random forest places greater emphasis on GIE, GIT, GFD, and GBD. The neural network ranks GST and GTFP first and second, with importance values of 0.1596 and 0.1493, respectively. SVR mainly emphasizes GIT and GST, with importance values of 0.2611 and 0.2467, respectively, while all three green technology variables have relatively low importance. Across the five models, GIT is the most stable core predictor, but the importance of GCT, GIE, GFD, GBD, and GST changes with model form. The roles of the green technology variables are more model-dependent, with GTFP making a relatively large contribution primarily in the neural network.

Across alternative combinations, the random forest model with 10 subvariables achieves a test-set R^2 of 0.5967. This represents only a modest improvement over the value of 0.5765 from the green finance subvariable model, although it is substantially higher than the value of 0.0601 from the model with three green technology subvariables. The predictive ability of the current joint model is therefore provided mainly by the green finance variables, with the three current-period green technology indicators contributing relatively little incremental information. Meanwhile, in the joint setting with two composite indices, SVR achieves a test-set R^2 of 0.7331 and an MAE of 0.7147, outperforming the model that directly includes the 10 subvariables. Thus, increasing the number of variables does not necessarily improve predictive performance. When correlations,

information redundancy, and measurement noise are present, composite indices may produce more stable predictions through dimensionality reduction and information integration.

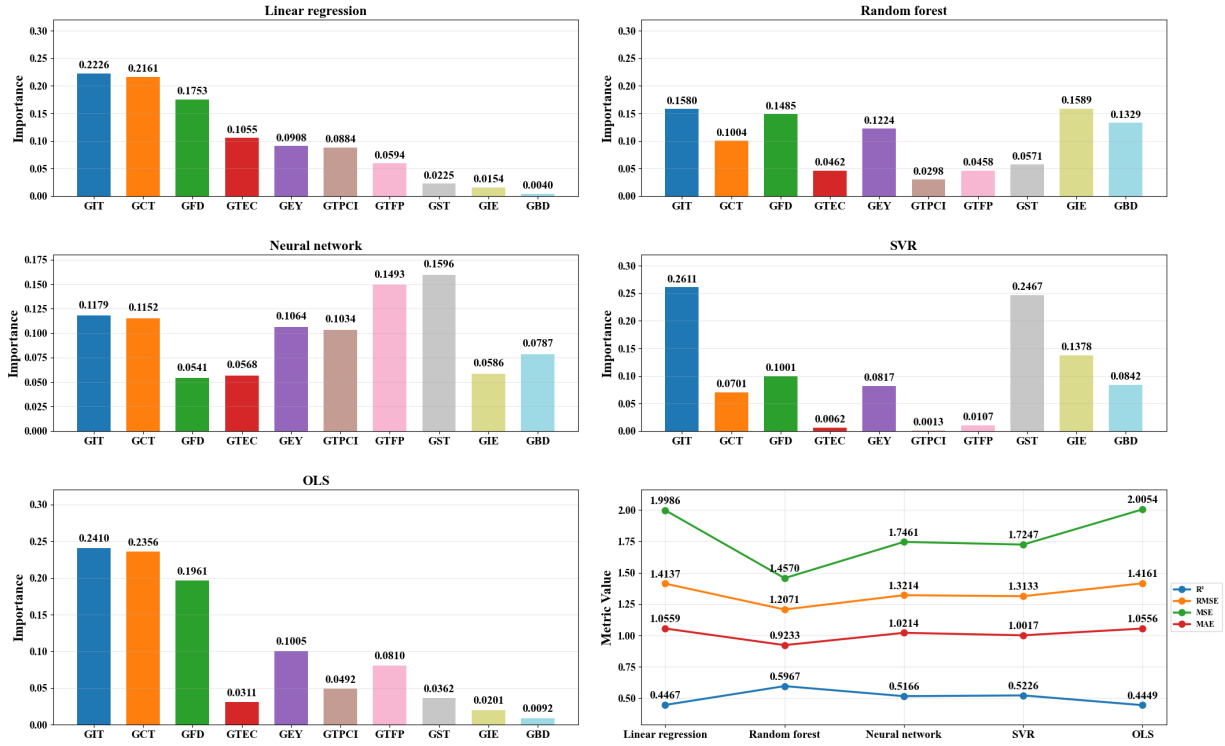


Figure 3. Prediction of Log_2GDP by green technology and green finance subvariables using five models.

4 Discussion

4.1 Distinct Roles of Green Technology and Green Finance and Their Economic Implications

Both green finance and green technology contain useful information about regional economic development, but their transmission pathways and temporal characteristics in relation to GDP differ. Green finance is linked to current economic activity mainly through resource allocation, project investment, and capital formation, and its role is therefore more readily reflected in current output. Green technology influences economic growth primarily through efficiency improvement, shifts in the production frontier, and technology adoption; its economic effects generally require research and development, transformation, diffusion, and large-scale application. The more direct relationship between green finance variables and current GDP therefore does not imply that the long-term role of green technology is weaker, but instead reflects differences in their effective time horizons.

GIT is the specific channel within the green finance system that has the most stable relationship with economic output. Investment in environmental pollution control can directly translate into infrastructure construction, equipment upgrades, engineering demand, and capital formation in green industries. Its economic transmission chain is therefore relatively short, making its output effect more readily observable than those of financial instruments that function primarily through intermediation or risk sharing. This

conclusion is consistent with the view of Shah and Wu (2025) that GIT promotes capital formation and economic activity. It must nevertheless be emphasized that variable importance represents a contribution to predictive information and cannot be interpreted directly as a causal effect.

The importance of different green finance instruments is strongly context-dependent. GCT, GBD, GIE, GFD, and GEY perform functions such as supplying finance, sharing risk, raising capital, and facilitating market pricing. Their effects depend on regional financial development, the quality of green projects, corporate financing constraints, and coordination among financial instruments. Expanding the scale of green finance therefore does not automatically promote economic growth. What matters is whether financial resources flow to green projects with genuine productive capacity and a sound technological foundation.

This study extends the existing literature on financing constraints by revealing heterogeneity among instruments within green finance. Prior studies generally hold that green finance promotes green technology and industrial development by reducing financing costs, broadening financing channels, and improving the availability of funds for green projects (Huang et al., 2022; Zhang et al., 2024; Ma and Zhuo, 2026; Shi et al., 2026; Sun et al., 2026). The present findings further indicate that different financial instruments do not follow identical mechanisms: channels that directly form capital inputs are relatively stable, whereas the roles of indirect financing and risk-management instruments depend more heavily on the institutional environment and the combination of instruments.

The relatively limited relationship between annual green technology indicators and current GDP reflects the lag in converting technological achievements into economic output. Annual changes in green technology efficiency and technological progress mainly describe relative changes in annual productive efficiency or the production frontier and are susceptible to short-term fluctuations, industrial structure, and differences in the base period. Technological research and development, however, generally translates into actual output only after knowledge accumulation, equipment upgrading, organizational learning, and market diffusion. Current technological changes may therefore not be immediately reflected in current GDP. Future research should examine lagged variables, technology stocks, or longer time horizons.

The relative prominence of GTFP in some nonlinear models indicates that composite efficiency information captures economic relationships more effectively than individual annual changes. GTFP simultaneously accounts for resource inputs, economic outputs, and environmental constraints and therefore provides a relatively comprehensive measure of regional green productive capacity, whereas changes in technological efficiency or technological progress describe only one component. However, because the green technology subvariable models have limited explanatory power overall, the high importance of a particular technology indicator indicates only a relative advantage among the available technology variables and does not imply a strong independent economic effect.

The superior performance of the composite technology indicator relative to the separate annual indicators demonstrates that indicator construction substantially affects the identification of the economic role of green technology. A composite index can reduce short-term fluctuations and measurement error in individual

indicators and more stably represent the overall state of regional green technological development. This finding does not conflict with studies that use the GML index, super-efficiency SBM models, and GTFP to measure green technology (Liu and Jiang, 2026). Rather, it indicates that the overall level of technological development may provide more stable information than annual changes for macroeconomic prediction.

The difference between this study and the existing literature lies primarily in its research objective and time horizon. Existing studies generally verify the contribution of green technology from the perspectives of high-quality economic development, resource allocation efficiency, and spatial spillovers (Wang et al., 2021; Shang et al., 2022; Jiang and Sun, 2024), whereas the present study focuses on the ability of green variables to predict differences in current GDP. The weak short-term predictive ability of the technology variables found here therefore does not negate the positive role of green technology in long-term growth and development quality. Instead, it suggests that technological effects may be difficult to identify fully through contemporaneous relationships with GDP.

Green economic growth is fundamentally constrained by both financial resources and the capacity to transform technology into production. In the absence of technological absorption and industrial commercialization capacity, GFD may remain at the level of financial expansion or be allocated to inefficient projects. Without long-term financial support, green technological achievements may remain at the research and experimental stage. Policy should therefore focus not merely on expanding the scale of green finance or raising technology indicators but on achieving an effective match among the supply of funds, technological capability, and the industrial foundation.

4.2 Complementarity Between Green Technology and Green Finance and Their Joint Information Gain

Including the Green Innovation Index and GFI simultaneously improves test-set performance across all five models. SVR achieves a test-set R^2 of 0.7331, while RMSE, MSE, and MAE decline to 0.9820, 0.9643, and 0.7147, respectively. Relative to the best green technology single-factor model and the best green finance single-factor model, the R^2 of the joint model increases by 0.1570 and 0.1720, respectively. These results indicate that green technology and green finance are not merely duplicate measures of the same regional development conditions but instead contain complementary information (Alam and Ullah, 2025; He et al., 2026).

In economic terms, green finance primarily changes access to funds, the cost of capital, and the direction of investment, whereas green technology primarily changes productive efficiency, patterns of energy use, and a region's potential productive capacity. They act, respectively, on the resource constraints and technological constraints of green projects. When a region has accumulated substantial technological capability but green firms face short financing maturities, insufficient collateral, and high research and development risks, its technological advantages may not be converted promptly into productive capacity and economic output. Conversely, when a region has abundant green financial resources but lacks commercially viable technology

projects, a skilled workforce, and technological absorption capacity, financial investment may be allocated inefficiently. A relatively complete transformation process involving research and development, technology adoption, equipment investment, industrialization, and expansion can emerge only when technological capability and financial supply are properly matched (He et al., 2026; Shi et al., 2026).

The improvement in the joint model therefore means more than a mechanical increase in fit after adding a variable; more importantly, it indicates that the two categories of green factors explain complementary aspects of GDP differences. Green finance can provide funding for technological research, development, and commercialization, whereas green technology determines whether that funding creates new productive capacity and sustained returns. A possible sequential process thus emerges: green finance improves the financing conditions of green projects, research and development and technology-adoption capacity subsequently increase, and technological progress then becomes associated with economic output through higher productivity and industrial restructuring.

It should nevertheless be emphasized that the machine-learning models used in this study identify predictive relationships and cannot directly establish a strict causal chain in which green finance promotes green technology and green technology subsequently promotes GDP. Accordingly, a more precise conclusion than the original claim that "this study identifies a continuous impact relationship" is that the results are consistent with a sequential mechanism linking financial support, technological accumulation, and economic output, and provide predictive empirical evidence for that mechanism. Its mediating effects and causal direction still require further verification using fixed-effects models, mediation models, lagged variables, instrumental variables, or policy-shock identification.

This conclusion extends prior studies that examine green technology and green finance separately. Research on green technology generally focuses on how technological innovation affects economic development through GTFP, industrial upgrading, and resource-use efficiency, whereas research on green finance focuses more on pathways involving financing constraints, talent agglomeration, environmental regulation, and capital allocation (Feng et al., 2024; Li et al., 2024; Zhang et al., 2026). These studies largely proceed within two relatively independent analytical frameworks and seldom test whether the information contained in green technology and green finance overlaps. The results from the joint model indicate that the two pathways cannot simply substitute for one another: financial resources cannot replace technological capability, and technological accumulation cannot fully substitute for capital support.

Differences in feature importance within the joint model provide further insight. In linear regression and OLS, the importance of GFI reaches 0.9751 and 0.9770, respectively, such that the models attribute almost the entire joint relationship to green finance. In random forest, neural network, and SVR, by contrast, importance is distributed approximately evenly, at 0.48-0.52, between GFI and CGTPI. This difference suggests that the relationship between green technology and green finance may not conform to a simple linear additive form.

Linear models can estimate only relationships with fixed directions and magnitudes on average across the sample. When two variables are correlated or exhibit threshold effects or interactions, their shared explanatory information may be concentrated in the variable with the stronger linear correlation with GDP. GFI may exhibit a stronger contemporaneous linear relationship and therefore absorb more of the joint explanatory power in linear models. Nonlinear models can identify structures in which the effect of financial support changes with the technological foundation and the effect of technological accumulation changes with financing conditions; they therefore assign more balanced contributions to the two factors.

In terms of policy implications, a larger linear coefficient cannot be interpreted simply as evidence that green finance is more important than green technology. Linear models reflect average linear associations, whereas nonlinear models may reveal conditional dependence and combined effects. In regions with a weak technological foundation, green finance may not rapidly generate high-quality output. Once technological accumulation reaches a certain level, the marginal effect of additional financial support may increase substantially. Similarly, when the financial supply is insufficient, the economic conversion rate of the stock of technological knowledge may be constrained. Such complementarity and threshold relationships are important reasons why joint nonlinear models outperform single-factor models.

4.3 Adaptive Relationships Between Green Factors and GDP

The variation in model performance across forms of indicator representation indicates that there is no single functional form applicable to all data structures in the relationship between green factors and GDP. Composite indices reduce dimensionality and smooth fluctuations in the indicators, whereas the original subvariables retain structural differences among financial instruments and technological dimensions. The fact that the two indicator systems correspond to different optimal algorithms means that model selection must proceed in tandem with variable construction; the same model should not be applied mechanically after the indicators change.

Composite indices are more suitable for identifying aggregate, smooth nonlinear relationships. The Green Innovation Index and GFI standardize and aggregate multiple original dimensions, reducing differences in scale, random fluctuations, and some collinearity and thereby making the feature space more continuous. By using a kernel function to capture nonlinear boundaries between low-dimensional variables while controlling model complexity, SVR is better suited to identifying stable relationships between the overall levels of green technology and green finance and GDP.

Original subvariables are more suitable for revealing instrument heterogeneity, threshold effects, and interactions among variables. GCT, investment, insurance, bonds, and funds differ in their funding sources, risk structures, and effective time horizons, while GTEC, GTPCI, and GTFP capture different aspects of productive capacity. Random forest can identify nonsmooth relationships and conditional combinations by partitioning the feature space and is therefore more adaptable when disaggregated variables are retained (Boateng et al., 2026).

The more stable predictive performance of the composite indices indicates that, in the current sample, the noise-reduction benefits of information aggregation outweigh the benefits of disaggregated information. Although the original variables provide richer information about mechanisms, they also introduce multicollinearity, measurement error, and short-term fluctuations, making it easier for a model to learn sample-specific relationships. Composite indices sacrifice some information about internal structure but improve the stability of the overall relationship and its out-of-sample applicability, reflecting a fundamental tradeoff between predictive accuracy and mechanism identification.

The dominance of green finance variables in the joint model with original indicators indicates that they have a more direct relationship with current economic activity. Different financial instruments can affect firm operations and project construction through investment formation, credit supply, risk sharing, and capital-market financing, whereas the economic effects of green technology variables generally take longer to materialize. The inclusion of annual technology indicators therefore does not substantially change the model's information structure. This reflects their limited marginal contribution to current prediction rather than an absence of long-term economic value from green technology.

The identification of key variables is model-dependent, and importance rankings cannot be interpreted independently of algorithmic structure. Linear models capture the average linear association between a variable and GDP; random forest emphasizes a variable's predictive contribution across different thresholds and sample partitions; and neural networks and SVR extract information through different nonlinear mappings. The different rankings produced by these models do not represent contradictory results but indicate that the roles of green factors may simultaneously involve average effects, conditional effects, and nonlinear interactions.

The stability of GIT across models makes it a relatively reliable candidate core variable. Compared with other financial instruments, GIT more directly links financial inputs to real projects, and its role therefore does not depend entirely on a particular functional form. By contrast, the importance of GIE, bonds, funds, and GST varies more widely, suggesting that these instruments are more likely to operate under specific market conditions, levels of risk, or policy combinations.

Composite and disaggregated indicators answer research questions at different levels. Composite indices primarily assess whether the overall development of green technology and green finance contains stable economic information, whereas disaggregated indicators identify the specific channels from which that information originates. The former are appropriate for aggregate evaluation and prediction, while the latter are suitable for mechanism analysis and policy design. The two types of indicators should therefore complement rather than replace one another.

The adaptive modeling framework extends the assumptions about functional form used in conventional econometric research. Existing studies mainly use fixed-effects, spatial econometric, mediation, and threshold models to test the average or spatial effects of green finance and green technology (Liu et al., 2022; Wang et al., 2022; Xue et al., 2022; Xu et al., 2023; Alfalih and Hadj, 2024; Hunjra et al., 2024; Li et al., 2024; Alfalih,

2025; Jia et al., 2025; Liang and Qiao, 2025; Qin et al., 2025; Lian et al., 2026). By comparing multiple models, this study shows that variable dimensionality and the method of aggregating indicators change the optimal functional form. Empirical research should therefore account for the relationship between indicator construction and algorithmic fit.

Machine-learning and econometric models should be regarded as complementary tools for discovering relationships and identifying causal effects. Machine learning is well suited to uncovering potential nonlinearities, thresholds, and interactions, but its feature importance primarily reflects predictive contribution and cannot directly establish a causal mechanism. Econometric models can further test statistical significance, endogeneity, and policy effects. A more robust research strategy would first use machine learning to identify potentially important relationships and then apply panel, spatial, or quasi-experimental methods for causal verification.

The policy implication of the adaptive relationship is that green policies must account for regional conditions and instrument matching. Local governments should not merely pursue simultaneous increases in the scale of green finance and green technology indicators. Instead, they should select appropriate policy combinations based on regional industrial structure, technological absorption capacity, financial-market maturity, and project risk. Green capital supply and technological transformation capacity must be properly matched for green finance and green technology to jointly generate stable economic output.

5 Conclusion

Based on 880 provincial-level observations, this study incorporates the Green Innovation Index, GFI, and their 11 subvariables into a unified analytical framework. It compares linear regression, OLS, random forest, neural network, and SVR on training and test sets and constructs an adaptive multimodel matching rule to investigate the independent relationships of green technology and green finance with GDP, the information gain from their joint inclusion, and differences in model fit.

Three conclusions emerge. First, both green finance and green technology are important correlates of regional economic development, but their principal roles differ. Green finance primarily reflects capital allocation and project investment, whereas green technology primarily reflects productive efficiency and overall technological capability. GIT is a relatively stable core financial variable, while the economic information in green technology is captured more strongly by its overall level of development than by short-term annual changes. Second, green technology and green finance are substantially complementary. Technological capability determines the efficiency with which GFD is transformed into output, while financial support influences the industrialization and large-scale application of green technology. Their synergy constitutes a dual condition for green economic development. Third, the relationship between green factors and GDP shows clear differences in model fit. Composite indices and original subvariables correspond to different optimal models. Adaptive matching across multiple models based on indicator level and feature

structure therefore captures the complex relationship between green development and economic growth more accurately than prespecifying a single model.

This study has three limitations. First, randomly dividing the training and test sets may disrupt the temporal dependence and regional correlation of provincial panel data. Future research should use temporal extrapolation, leave-one-region-out validation, and spatial cross-validation. Second, on the basis of the relationships discovered by multiple models, future studies should introduce province and year fixed effects, lagged variables, instrumental variables, policy shocks, spatial econometric models, or mediation models and further examine the causal mechanisms through which green technology and green finance affect GDP.

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